BUS 104: Practice Set — Linear Programming (LP)

Problems and Solutions

Problem 2-40: Feed 'n' Ship Ranch

The Feed 'n' Ship Ranch fattens cattle for local farmers and ships them to meat markets in Kansas City and Omaha. The owners of the ranch seek to determine the amounts of cattle feed to buy so that minimum nutritional standards are satisfied and at the same time total feed costs are minimized. The feed mix used can be made up of three grains that contain the following nutrients per pound of feed:

	Nutrient (Ounces per Pound of Feed)			
Feed	**A**	**B**	**C**	**D**
Stock X	3	2	1	6
Stock Y	2	3	0	8
Stock Z	4	1	2	4

The cost per pound of stocks X, Y, and Z are \$3, \$4, and \$2.25, respectively. The minimum requirement per animal per month is 4 pounds of nutrient A, 5 pounds of nutrient B, 1 pound of nutrient C, and 8 pounds of nutrient D.

The ranch faces one additional restriction: It can obtain only 500 pounds of stock Z per month from the feed supplier, regardless of its need. Because there are usually 100 cattle at the Feed 'n' Ship Ranch at any given time, this means that no more than 5 pounds of stock Z can be counted on for use in the feed of each animal per month. Formulate this problem as a linear program and solve it by using Excel.

Solution to Problem 2-40

1. Decision Variables and Unit Conversions

Let X, Y, Z = number of pounds of Stock X, Stock Y, and Stock Z purchased per cow each month. Because the nutrient content in the table is given in *ounces* per pound of feed while the monthly requirements per cow are stated in *pounds*, we convert the monthly minimum requirements to ounces (1 lb = 16 oz):

- **Nutrient A:** 4 lbs $\times$ 16 oz/lb = 64 oz; **Nutrient B:** 5 lbs $\times$ 16 oz/lb = 80 oz
- **Nutrient C:** 1 lb $\times$ 16 oz/lb = 16 oz; **Nutrient D:** 8 lbs $\times$ 16 oz/lb = 128 oz

2. Linear Programming Formulation

$$\text{Minimize Cost} = \$3X + \$4Y + \$2.25Z$$

subject to:

$$3X + 2Y + 4Z \geq 64 \text{ (Nutrient A, oz)}, \qquad 2X + 3Y + 1Z \geq 80 \text{ (Nutrient B, oz)}$$
$$1X + 0Y + 2Z \geq 16 \text{ (Nutrient C, oz)}, \qquad 6X + 8Y + 4Z \geq 128 \text{ (Nutrient D, oz)}$$
$$Z \leq 5 \text{ (Stock Z limit, lbs)}, \qquad X, Y, Z \geq 0 \text{ (Non-negativity)}$$

3. Optimal Solution (see `P2-40.xls`): Stock X (X): 16 lbs, **Stock Y (Y):** 16 lbs, **Stock Z (Z):** 0 lbs; **Minimum Cost:** \$**112.00** per cow/month.

Spreadsheet Row	**Stock X (X)**	**Stock Y (Y)**	**Stock Z (Z)**	**LHS**	**Sign**	**RHS**
# of pounds	16.00	16.00	0.00			
Cost (\$)	3.00	4.00	2.25	**\$112.00**	*(Objective)*	
Ingr. A required	3	2	4	80	$\geq$	64
Ingr. B required	2	3	1	80	$\geq$	80
Ingr. C required	1	0	2	16	$\geq$	16
Ingr. D required	6	8	4	224	$\geq$	128
Stock Z max	0	0	1	0	$\leq$	5

Note: In the 1st edition of the textbook (shown on Sheet 2 of `P2-40.xls`), the unit costs were $\$2X + \$4Y + \$2.50Z$, which yields $X = 40$ lbs, $Y = 0$ lbs, $Z = 0$ lbs, and a minimum cost of \$80.00.

Problem 2-41: Weinberger Electronics Corporation

The Weinberger Electronics Corporation manufactures four highly technical products that it supplies to aerospace firms that hold NASA contracts. Each of the products must pass through the following departments before they are shipped: wiring, drilling, assembly, and inspection. The time requirement (in hours) for each unit produced, the available time in each department each month, minimum production levels for each product, and unit profits for each product are summarized in the following table:

	Department					
Product	**Wiring**	**Drilling**	**Assembly**	**Inspection**	**Unit Profit**	**Min. Needed**
XJ201	0.5	0.3	0.2	0.5	\$9	150
XM897	1.5	1.0	4.0	1.0	\$12	100
TR29	1.5	2.0	1.0	0.5	\$15	300
BR788	1.0	3.0	2.0	0.5	\$11	400
Capacity (Hours)	15,000	17,000	10,000*	12,000		

*Note: The spreadsheet file `P2-41.xls` uses 26,000 hours for assembly capacity (from an earlier printing). Because only 8,080 hours of assembly time are used at the optimum, both capacities yield the exact same optimal solution.

The production manager has the responsibility of specifying production levels for each product for the coming month. Help him by formulating the problem and solving it by using Excel.

Solution to Problem 2-41

1. Decision Variables

Let X_1, X_2, X_3, X_4 = number of units of XJ201, XM897, TR29, and BR788 produced per month.

2. Linear Programming Formulation

$$\text{Maximize Profit} = \$9X_1 + \$12X_2 + \$15X_3 + \$11X_4$$

subject to:

$$0.5X_1 + 1.5X_2 + 1.5X_3 + 1.0X_4 \leq 15{,}000 \quad \text{(Wiring hours available)}$$
$$0.3X_1 + 1.0X_2 + 2.0X_3 + 3.0X_4 \leq 17{,}000 \quad \text{(Drilling hours available)}$$
$$0.2X_1 + 4.0X_2 + 1.0X_3 + 2.0X_4 \leq 26{,}000 \quad \text{(Assembly hours available; or 10,000)}$$
$$0.5X_1 + 1.0X_2 + 0.5X_3 + 0.5X_4 \leq 12{,}000 \quad \text{(Inspection hours available)}$$
$$X_1 \geq 150, \quad X_2 \geq 100, \quad X_3 \geq 300, \quad X_4 \geq 400 \quad \text{(Minimum production requirements)}$$
$$X_1,\ X_2,\ X_3,\ X_4 \geq 0 \quad \text{(Non-negativity)}$$

3. Optimal Solution (see `P2-41.xls`)

- **XJ201 (X_1):** 20,650 units, **XM897 (X_2):** 100 units, **TR29 (X_3):** 2,750 units, **BR788 (X_4):** 400 units
- **Maximum Total Profit:** \$232,700.00

Spreadsheet Row	**XJ201**	**XM897**	**TR29**	**BR788**	**LHS**	**Sign**	**RHS**
# of units	20,650	100	2,750	400			
Profit (\$)	9	12	15	11	**\$232,700**	*(Objective)*	
Wiring time	0.5	1.5	1.5	1.0	15,000	≤	15,000
Drilling time	0.3	1.0	2.0	3.0	12,995	≤	17,000
Assembly time	0.2	4.0	1.0	2.0	8,080	≤	26,000
Inspection time	0.5	1.0	0.5	0.5	12,000	≤	12,000
Minimum XJ201	1	0	0	0	20,650	≥	150
Minimum XM897	0	1	0	0	100	≥	100
Minimum TR29	0	0	1	0	2,750	≥	300
Minimum BR788	0	0	0	1	400	≥	400

Problem 2-43: Canal Street Party Mix Company (CSPM)

The Canal Street Party Mix Company (CSPM) packages and sells three different one-pound canned party mixes: Plain Nuts, Mixed Nuts, and Premium Mix. Plain Nuts sells for \$2.25 per can, Mixed Nuts sells for \$3.37, and Premium Nuts sells for \$6.49 per can. A can of Plain Nuts contains 0.8 pound peanuts and 0.2 pound cashews. A can of Mixed Nuts consists of 0.5 pound peanuts, 0.3 pound cashews, 0.1 pound almonds, and 0.1 pound walnuts. A can of Premium Nuts is made up of 0.3 pound cashews, 0.4 pound almonds, and 0.4 pound walnuts. CSPM has on hand 500 pounds of peanuts, 225 pounds of cashews, 100 pounds of almonds, and 80 pounds of walnuts. Past demand indicates that customers purchase at least twice as many cans of Plain Nuts as Premium Nuts. What should CSPM do if it would like to produce the different products in a way that will maximize total revenue?

Solution to Problem 2-43

1. Decision Variables

Let:

- P = number of cans of Plain Nuts to produce
- M = number of cans of Mixed Nuts to produce
- R = number of cans of Premium Nuts to produce

2. Linear Programming Formulation

$$\text{Maximize Revenue} = \$2.25P + \$3.37M + \$6.49R$$

subject to:

$$\begin{aligned}
0.8P + 0.5M &\leq 500 && \text{(Peanuts available, in lbs)} \\
0.2P + 0.3M + 0.3R &\leq 225 && \text{(Cashews available, in lbs)} \\
0.1M + 0.4R &\leq 100 && \text{(Almonds available, in lbs)} \\
0.1M + 0.4R &\leq 80 && \text{(Walnuts available, in lbs)} \\
P &\geq 2R && \text{(Plain vs. Premium demand requirement)} \\
P,\ M,\ R &\geq 0 && \text{(Non-negativity)}
\end{aligned}$$

3. Optimal Solution (see `P2-43.xls`)

- **Plain Nuts (P):** 375 cans, **Mixed Nuts (M):** 400 cans, **Premium Nuts (R):** 100 cans
- **Maximum Total Revenue:** \$2,840.75

Spreadsheet Row	**Plain (P)**	**Mixed (M)**	**Premium (R)**	**LHS**	**Sign**	**RHS**
Number of cans	375.00	400.00	100.00			
Revenue (\$)	2.25	3.37	6.49	**\$2,840.75**	*(Objective)*	
Peanuts	0.8	0.5	0.0	500.00	$\leq$	500.00
Cashews	0.2	0.3	0.3	225.00	$\leq$	225.00
Almonds	0.0	0.1	0.4	80.00	$\leq$	100.00
Walnuts	0.0	0.1	0.4	80.00	$\leq$	80.00
Plain vs. Premium ($P \geq 2R$)	1.0	0.0	0.0	375.00	$\geq$	200.00 ($= 2R$)

Problem 2-44: Railroad Stock Portfolio

An investor is considering three different railroad stocks to complement his portfolio: B & O Railroad, Short Line Railroad, and Reading Railroad. His broker has given him the following information:

	Stock ($)		
Factor	**B & O**	**Short Line**	**Reading**
Short-term growth per $ invested	0.39	0.26	0.42
Intermediate growth per $ invested	1.59	1.70	1.45*
Dividend rate	8%	4%	6%

*Note: The spreadsheet file `P2-44.xls` uses 1.55 for Reading's intermediate-term growth per dollar invested. Under both 1.45 and 1.55, no funds are invested in Reading ($R = \$0$), so the optimal portfolio is identical.

The investor's criteria are as follows: (1) Investments should yield short-term growth of at least $1,000, (2) investments should yield intermediate-term growth of at least $6,000, and (3) dividends should be at least $250 per year. Determine the least amount the investor can invest and how that investment should be allocated among the three stocks.

Solution to Problem 2-44

1. Decision Variables

Let:

- B = dollars invested in B & O Railroad
- S = dollars invested in Short Line Railroad
- R = dollars invested in Reading Railroad

2. Linear Programming Formulation

$$\text{Minimize Total Investment} = B + S + R$$

subject to:

$$\begin{aligned}
0.39B + 0.26S + 0.42R &\geq \$1{,}000 \quad \text{(Short-term growth)} \\
1.59B + 1.70S + 1.55R &\geq \$6{,}000 \quad \text{(Intermediate-term growth; or } 1.45R\text{)} \\
0.08B + 0.04S + 0.06R &\geq \$250 \quad \text{(Dividend income)} \\
B,\ S,\ R &\geq 0 \quad \text{(Non-negativity)}
\end{aligned}$$

3. Optimal Solution (see `P2-44.xls`)

- **B & O Railroad (B):** \$2,555.25, **Short Line Railroad (S):** \$1,139.50, **Reading Railroad (R):** \$0.00
- **Minimum Total Investment:** \$3,694.75

Spreadsheet Row	**B & O (B)**	**Short (S)**	**Reading (R)**	**LHS**	**Sign**	**RHS**
$ invested	\$2,555.25	\$1,139.50	\$0.00			
Investment	1	1	1	**\$3,694.75**	*(Objective)*	
Short-term growth	0.39	0.26	0.42	\$1,292.82	≥	\$1,000.00
Intermediate growth	1.59	1.70	1.55	\$6,000.00	≥	\$6,000.00
Dividend income	0.08	0.04	0.06	\$250.00	≥	\$250.00

Problem 3-2: Jose Garcia's Backpack Company

Jose Garcia, a budding entrepreneur, has set up a small backpack company and wants to distribute the backpacks exclusively through the local bookstore. There are four models of backpacks: Canvas, Plastic, Nylon, and Leather. The bookstore wants no fewer than 15 and no more than 40 of any one model. The bookstore has agreed to pay Jose \$35.50 for each canvas backpack, \$39.50 for each plastic backpack, \$42.50 for each nylon backpack, and \$69.50 for each leather backpack he can deliver by the end of next week.

Jose has three student friends with sewing machines who have agreed to do the sewing for him. One of the students can work on either canvas or plastic, can complete a backpack in 1.5 hours, and will charge \$7 per hour to do the work. She can commit to 90 hours of work in the next week. Another student can work on the nylon fabric. He can sew a bag in 1.7 hours, will charge \$8 per hour to work, and can work 42.5 hours in the next week. The third student has the ability to sew leather. He can complete a book bag in 1.9 hours, will charge \$9 per hour to work, and can work 80 hours during the next week. The following table provides additional information about each backpack:

	Canvas	**Plastic**	**Nylon**	**Leather**
Square yards per pack	2.25	2.40	2.10	2.60
Square yards available	200	350	700	550
Cost per square yard	\$4.50	\$4.25	\$7.65	\$9.45

What is the best combination of backpacks for Jose to plan to deliver to the bookstore?

Solution to Problem 3-2

1. Decision Variables and Unit Profits

Let C, P, N, L denote the number of Canvas, Plastic, Nylon, and Leather backpacks produced. Using unit profit = Revenue − Material Cost − Labor Cost:

- **Canvas (C):** \$35.50 − \$10.125 − \$10.50 = **\$14.875**; **Plastic (P):** \$39.50 − \$10.20 − \$10.50 = **\$18.80**
- **Nylon (N):** \$42.50 − \$16.065 − \$13.60 = **\$12.835**; **Leather (L):** \$69.50 − \$24.57 − \$17.10 = **\$27.83**

2. Linear Programming Formulation

$$\text{Maximize Profit} = \$14.875C + \$18.80P + \$12.835N + \$27.83L$$

subject to:

$$2.25C \le 200,\ \ 2.40P \le 350,\ \ 2.10N \le 700,\ \ 2.60L \le 550 \quad \text{(Material available, sq yds)}$$

$$1.5C + 1.5P \le 90,\ \ 1.7N \le 42.5,\ \ 1.9L \le 80,\ \ 15 \le C, P, N, L \le 40 \quad \text{(Labor hrs \& order limits)}$$

3. Optimal Solution (see `P3-2.xls`)

- **Canvas (C):** 20, **Plastic (P):** 40, **Nylon (N):** 25, **Leather (L):** 40 backpacks; **Max Profit:** **\$2,483.58**

Spreadsheet Row	**Canvas (C)**	**Plastic (P)**	**Nylon (N)**	**Leather (L)**	**LHS**	**Sign**	**RHS**
Solution value	20.0	40.0	25.0	40.0			
Profit (\$)	14.875	18.800	12.835	27.830	**\$2,483.58**	*(Objective)*	
Canvas / Plastic avail.	2.25	2.40	0	0	45.0, 96.0	≤	200, 350
Nylon / Leather avail.	0	0	2.10	2.60	52.5, 104.0	≤	700, 550
Canvas & plastic labor	1.5	1.5	0	0	90.0	≤	90.0
Nylon / Leather labor	0	0	1.7	1.9	42.5, 76.0	≤	42.5, 80.0
Min / Max per model	1	1	1	1	20, 40, 25, 40	∈	[15, 40]

Problem 3-5: Morrison Elementary School Fundraiser

Students at Morrison Elementary School are trying to raise funds to send their team to the national "Odyssey of the Mind" competition in Denver, Colorado, next month. A local pizza chain has agreed to sell them pizzas at a discount, which the students can then resell to families in the local community for a profit. It is expected that of the 500 families in the community, at most 70% will buy pizzas. Based on a survey of their personal preferences, the students believe that they should order no more than 120 cheese pizzas, no more than 150 pepperoni pizzas, and no more than 100 vegetarian pizzas. They also want to make sure that at least 20% of the total pizzas are cheese and at least 50% of the pizzas are pepperoni. They make a profit of \$1.45, \$1.75, and \$1.98, respectively, for each cheese, pepperoni, and vegetarian pizza they resell. How many pizzas of each type should they buy?

Solution to Problem 3-5

1. Decision Variables

Let:

- C = number of cheese pizzas to order
- P = number of pepperoni pizzas to order
- V = number of vegetarian pizzas to order

2. Linear Programming Formulation

At most 70% of the 500 families will buy pizzas, so total pizza demand is at most $0.70 \times 500 = 350$ pizzas.

$$\text{Maximize Profit} = \$1.45C + \$1.75P + \$1.98V$$

subject to:

$$C + P + V \leq 350 \quad \text{(Maximum total pizzas sold)}$$

$$C \leq 120 \quad \text{(Maximum cheese pizzas)}$$

$$P \leq 150 \quad \text{(Maximum pepperoni pizzas)}$$

$$V \leq 100 \quad \text{(Maximum vegetarian pizzas)}$$

$$C \geq 0.20(C + P + V) \iff 0.80C - 0.20P - 0.20V \geq 0 \quad \text{(Minimum 20\% cheese)}$$

$$P \geq 0.50(C + P + V) \iff -0.50C + 0.50P - 0.50V \geq 0 \quad \text{(Minimum 50\% pepperoni)}$$

$$C, P, V \geq 0 \quad \text{(Non-negativity)}$$

3. Optimal Solution (see `P3-5.xls`)

- **Cheese (C):** 60 pizzas, **Pepperoni (P):** 150 pizzas, **Vegetarian (V):** 90 pizzas
- **Maximum Total Profit:** \$527.70 (with 300 total pizzas ordered)

Spreadsheet Row	**Cheese (C)**	**Pepperoni (P)**	**Veggie (V)**	**LHS**	**Sign**	**RHS**
Solution value	60.0	150.0	90.0			
Profit (\$)	1.45	1.75	1.98	**\$527.70**	*(Objective)*	
Total pizzas	1	1	1	300.0	$\leq$	350.0
Max cheese	1	0	0	60.0	$\leq$	120.0
Max pepperoni	0	1	0	150.0	$\leq$	150.0
Max veggie	0	0	1	90.0	$\leq$	100.0
Min cheese ($\geq$ 20%)	1	0	0	60.0	$\geq$	60.0 (= 0.2 Total)
Min pepperoni ($\geq$ 50%)	0	1	0	150.0	$\geq$	150.0 (= 0.5 Total)

Problem 3-6: Bob Bell's 40th Birthday Party

Bob Bell's 40th birthday party promises to be the social event of the year in Cookeville. To prepare, Bob stocks up on the following liquors:

Liquor	Available Ounces
Bourbon	64
Brandy	38
Vodka	70
Dry vermouth	30
Sweet vermouth	40

Bob decides to mix four drinks for the party: Chaunceys, Sweet Italians, bourbon on the rocks, and Russian martinis. A Chauncey consists of 25% bourbon, 25% brandy, 25% vodka, and 25% sweet vermouth. A Sweet Italian contains 25% brandy, 25% dry vermouth, and 50% sweet vermouth. Bourbon on the rocks contains only bourbon. Finally, a Russian martini consists of 75% vodka and 25% dry vermouth. Each drink contains 4 fluid ounces.

Bob's objective is to mix these ingredients in such a way as to make the largest possible number of drinks in advance. Formulate Bob's problem as an LP model and solve it by using Excel.

Solution to Problem 3-6

1. Decision Variables

Let:

- C = number of Chaunceys mixed, S = number of Sweet Italians mixed
- B = number of Bourbon on the rocks mixed, R = number of Russian martinis mixed

2. Linear Programming Formulation

Since each drink contains 4 fluid ounces, the liquor percentages translate directly into ounces per drink (25% = 1 oz, 50% = 2 oz, 75% = 3 oz, 100% = 4 oz):

$$\text{Maximize Total Drinks} = C + S + B + R$$

subject to:

$1C + 4B \leq 64$ (Bourbon, oz), $1C + 1S \leq 38$ (Brandy, oz), $1C + 3R \leq 70$ (Vodka, oz)

$1S + 1R \leq 30$ (Dry vermouth, oz), $1C + 2S \leq 40$ (Sweet vermouth, oz), $C, S, B, R \geq 0$

3. Optimal Solution (see `P3-6.xls`)

- **Chaunceys (C):** 16, **Sweet Italians (S):** 12, **Bourbon on the rocks (B):** 12, **Russian martinis (R):** 18
- **Maximum Total Drinks:** 58 drinks

Spreadsheet Row	Chauncey	Sweet Ital.	Bourbon	Russian Mart.	LHS	Sign	RHS
Number of drinks	16.0	12.0	12.0	18.0			
Obj. coefficient	1	1	1	1	**58.0**	*(Objective)*	
Bourbon (oz)	1	0	4	0	64.0	≤	64.0
Brandy (oz)	1	1	0	0	28.0	≤	38.0
Vodka (oz)	1	0	0	3	70.0	≤	70.0
Dry vermouth (oz)	0	1	0	1	30.0	≤	30.0
Sweet vermouth (oz)	1	2	0	0	40.0	≤	40.0

Problem 3-9 (Extra Problem): Diversey Paint and Supply

The advertising director for Diversey Paint and Supply, a chain of four retail stores on Chicago's north side, is considering three media possibilities: (1) ads in the Sunday newspaper, (2) ads in a local trade magazine that is distributed free to all houses in the city and northwest suburbs, and (3) ads on a local TV station. The stores are expanding their lines of do-it-yourself tools, and the advertising director is interested in a total new-customer exposure level of at least 40% within the city and 60% in the northwest suburbs.

Each TV ad has a new-customer exposure level of 5% in the city and 3% in the northwest suburbs. The newspaper ads have corresponding exposure levels per ad of 3.5% and 3%, while the trade magazine has corresponding exposure levels per ad of 0.5% and 1%. The relevant costs are \$1,000 per newspaper ad, \$300 per trade magazine ad, and \$2,000 per TV ad. So that all three types of media are used, Diversey would like to ensure that no single medium consumes more than 45% of the total amount spent.

Diversey would like to select the least costly advertising strategy that would meet desired exposure levels. Formulate the problem using LP and solve it by using Excel.

Solution to Problem 3-9

1. Decision Variables

Let X_i = number of ads placed in medium i ($i = 1, 2, 3$):

- X_1 = number of Sunday newspaper ads
- X_2 = number of local trade magazine ads
- X_3 = number of local TV ads

2. Linear Programming Formulation

$$\text{Minimize Cost} = \$1{,}000X_1 + \$300X_2 + \$2{,}000X_3$$

subject to:

$$3.5X_1 + 0.5X_2 + 5X_3 \geq 40 \quad \text{(Exposure in the city, in \%)}$$
$$3.0X_1 + 1.0X_2 + 3X_3 \geq 60 \quad \text{(Exposure in the northwest suburbs, in \%)}$$

and the media budget share constraints (no single medium consumes more than 45% of total spending):

$$1{,}000X_1 \leq 0.45(1{,}000X_1 + 300X_2 + 2{,}000X_3) \iff -550X_1 + 135X_2 + 900X_3 \geq 0$$
$$300X_2 \leq 0.45(1{,}000X_1 + 300X_2 + 2{,}000X_3) \iff 450X_1 - 165X_2 + 900X_3 \geq 0$$
$$2{,}000X_3 \leq 0.45(1{,}000X_1 + 300X_2 + 2{,}000X_3) \iff 450X_1 + 135X_2 - 1{,}100X_3 \geq 0$$
$$X_1,\ X_2,\ X_3 \geq 0 \quad \text{(Non-negativity)}$$

3. Optimal Solution

- **Ads in newspaper (X_1):** 9
- **Ads in trade magazine (X_2):** 30
- **Ads on TV (X_3):** 1
- **Minimum Total Cost:** \$20,000.00

Problem 3-18: Y.S. Chang Restaurant Staffing

The famous Y.S. Chang Restaurant is open 24 hours a day. Waiters and busboys report for duty at 3 A.M., 7 A.M., 11 A.M., 3 P.M., 7 P.M., or 11 P.M., and each works an 8-hour shift. The following table shows the minimum number of workers needed during the six periods into which the day is divided:

Period	Time	Workers Required
1	3 A.M.–7 A.M.	3
2	7 A.M.–11 A.M.	12
3	11 A.M.–3 P.M.	16
4	3 P.M.–7 P.M.	9
5	7 P.M.–11 P.M.	11
6	11 P.M.–3 A.M.	4

How should Chang schedule his workers so that the total staff required for one day's operation is minimized?

Solution to Problem 3-18

1. Decision Variables

Let X_i = number of workers starting their 8-hour shift at the beginning of Period i ($i = 1, 2, 3, 4, 5, 6$):

- X_1 = start at 3 A.M., X_2 = start at 7 A.M., X_3 = start at 11 A.M., X_4 = start at 3 P.M., X_5 = start at 7 P.M., X_6 = start at 11 P.M.

2. Linear Programming Formulation

Because each shift lasts 8 hours (covering two consecutive 4-hour periods), workers on duty during Period i are those who started in Period $i-1$ plus those who started in Period i (wrapping around from Period 6 to Period 1):

$$\text{Minimize Total Staff} = X_1 + X_2 + X_3 + X_4 + X_5 + X_6$$

subject to:

$X_1 + X_6 \geq 3$ (3 A.M.–7 A.M.), $X_1 + X_2 \geq 12$ (7 A.M.–11 A.M.), $X_2 + X_3 \geq 16$ (11 A.M.–3 P.M.)

$X_3 + X_4 \geq 9$ (3 P.M.–7 P.M.), $X_4 + X_5 \geq 11$ (7 P.M.–11 P.M.), $X_5 + X_6 \geq 4$ (11 P.M.–3 A.M.)

with $X_1, X_2, X_3, X_4, X_5, X_6 \geq 0$.

3. Optimal Solution (see `P3-18.xls`)

- **Start at 3 a.m.** (X_1)**:** 0, **Start at 7 a.m.** (X_2)**:** 12, **Start at 11 a.m.** (X_3)**:** 4
- **Start at 3 p.m.** (X_4)**:** 10, **Start at 7 p.m.** (X_5)**:** 1, **Start at 11 p.m.** (X_6)**:** 3
- **Minimum Total Staff Size:** 30 workers *(Note: Multiple optimal solutions exist.)*

Spreadsheet Row	3am	7am	11am	3pm	7pm	11pm	LHS	Sign	RHS
Number of waiters	0.0	12.0	4.0	10.0	1.0	3.0			
Objective	1	1	1	1	1	1	**30.0**	*(Objective)*	
3am–7am needs	1	0	0	0	0	1	3.0	≥	3.0
7am–11am needs	1	1	0	0	0	0	12.0	≥	12.0
11am–3pm needs	0	1	1	0	0	0	16.0	≥	16.0
3pm–7pm needs	0	0	1	1	0	0	14.0	≥	9.0
7pm–11pm needs	0	0	0	1	1	0	11.0	≥	11.0
11pm–3am needs	0	0	0	0	1	1	4.0	≥	4.0

Problem 3-20: St. Jude's Hospital Lab Technicians

Elaine Jones, lab supervisor at St. Jude's Hospital, is considering scheduling her techs to work 12-hour shifts instead of the usual 8-hour shifts. Instead of working five 8-hour days, the lab techs would work three days on, four days off, followed by four days on and three days off, for a total of 84 hours every two weeks. Elaine feels this would be an improved schedule and would serve as a good incentive because it would give the employees more consecutive days off.

Elaine has decided that because the peak demand times in the hospital appear to be between 5 A.M. and 7 A.M., and between 5 P.M. and 7 P.M., she will arrange four 12-hour shifts according to the following table:

Shifts	Work Times	Pay Rate/Week
A and A (alt)	5 A.M.–5 P.M.	\$756
B and B (alt)	7 A.M.–7 P.M.	\$840
C and C (alt)	5 P.M.–5 A.M.	\$882
D and D (alt)	7 P.M.–7 A.M.	\$924

The shift pay differentials are based on the most and least desirable times to begin and end work. In any one week, techs on shift A might work Sunday through Tuesday, while techs on shift A (alt) would work at the same times, but on Wednesday through Saturday. In the following week, techs on shift A would work Sunday through Wednesday, while techs on shift A (alt) would work the corresponding Thursday through Saturday. Therefore, the same number of techs would be scheduled for shift A as for shift A (alt). The requirements for lab techs during the 24-hour day are as shown in the following table:

	5 a.m.–7 a.m.	7 a.m.–5 p.m.	5 p.m.–7 p.m.	7 p.m.–5 a.m.
Lab techs needed	10	6	12	8

What is the most economical schedule for the lab techs?

Solution to Problem 3-20

1. Decision Variables

Let X_1, X_2, X_3, X_4 be the number of technicians assigned to each crew of shift pairs A/A(alt), B/B(alt), C/C(alt), and D/D(alt), respectively. Because each shift schedule requires two equal-sized crews (primary and alternate), assigning X_i technicians on duty per day requires paying $2X_i$ technicians each week.

2. Linear Programming Formulation

$$\text{Minimize Weekly Payroll} = 2(\$756X_1 + \$840X_2 + \$882X_3 + \$924X_4) = \$1{,}512X_1 + \$1{,}680X_2 + \$1{,}764X_3 + \$1{,}848X_4$$

subject to:

$X_1 + X_4 \geq 10$ (5 A.M.–7 A.M. needs), $X_1 + X_2 \geq 6$ (7 A.M.–5 P.M. needs)

$X_2 + X_3 \geq 12$ (5 P.M.–7 P.M. needs), $X_3 + X_4 \geq 8$ (7 P.M.–5 A.M. needs), $X_1, X_2, X_3, X_4 \geq 0$

3. Optimal Solution (see `P3-20.xls`)

- **Shifts A & A (alt) (X_1):** 10, **Shifts B & B (alt) (X_2):** 4, **Shifts C & C (alt) (X_3):** 8, **Shifts D & D (alt) (X_4):** 0
- **Minimum Weekly Payroll:** \$35,952.00

Spreadsheet Row	A & A(alt)	B & B(alt)	C & C(alt)	D & D(alt)	LHS	Sign	RHS
Number assigned	10.0	4.0	8.0	0.0			
Pay (\$)	1,512.00	1,680.00	1,764.00	1,848.00	**\$35,952.00**	*(Objective)*	
5am–7am needs	1	0	0	1	10.0	≥	10.0
7am–5pm needs	1	1	0	0	14.0	≥	6.0
5pm–7pm needs	0	1	1	0	12.0	≥	12.0
7pm–5am needs	0	0	1	1	8.0	≥	8.0

Problem 3-21: San Diego Airline Ticket Counter Staffing

An airline with operations in San Diego, California, must staff its ticket counters inside the airport. Ticket attendants work six-hour shifts at the counter. There are two types of agents—those who speak English as a first language and those who are fully bilingual (English and Spanish). The requirements for the number of agents are a function of the numbers of people expected to pass through the airline's ticket counters during various hours. The airline believes that the need for agents between the hours of 6 A.M. and 9 P.M. are as in the table below:

	6 a.m.–9 a.m.	**9 a.m.–Noon**	**Noon–3 p.m.**	**3 p.m.–6 p.m.**	**6 p.m.–9 p.m.**
Ticket agents needed	12	20	16	24	12

Agents begin work either at 6 A.M., 9 A.M., noon, or 3 P.M. The shifts are designated as shifts A, B, C, and D, respectively. It is the policy of the airline that at least half of the agents needed in any time period will speak English as the first language. Further, at least one-quarter of the agents needed in any time period should be fully bilingual.

(a) How many and what type of agents should be hired for each shift to meet the language and staffing requirements for the airline?

(b) What is the optimal hiring plan if English-speaking agents are paid \$19 per hour and bilingual agents are paid \$23 per hour? Does the total number of agents needed change from that computed in part (a)?

Solution to Problem 3-21

1. Decision Variables and Formulations

Let E_A, E_B, E_C, E_D and B_A, B_B, B_C, B_D be the number of English-speaking and bilingual agents starting on 6-hour Shifts A (6 A.M.), B (9 A.M.), C (noon), and D (3 P.M.), respectively.

- **Part (a) Objective:** Minimize Total Agents $= E_A + E_B + E_C + E_D + B_A + B_B + B_C + B_D$
- **Part (b) Objective:** Minimize Hourly Wage Cost $= \$19(E_A + E_B + E_C + E_D) + \$23(B_A + B_B + B_C + B_D)$

subject to (for both parts):

$$\begin{aligned}
\textbf{Total Needed:}\quad & E_A + B_A \geq 12,\ \ E_A + E_B + B_A + B_B \geq 20,\ \ E_B + E_C + B_B + B_C \geq 16,\\
& E_C + E_D + B_C + B_D \geq 24,\ \ E_D + B_D \geq 12\\
\textbf{English } (\geq 50\%)\textbf{:}\quad & E_A \geq 6,\ \ E_A + E_B \geq 10,\ \ E_B + E_C \geq 8,\ \ E_C + E_D \geq 12,\ \ E_D \geq 6\\
\textbf{Bilingual } (\geq 25\%)\textbf{:}\quad & B_A \geq 3,\ \ B_A + B_B \geq 5,\ \ B_B + B_C \geq 4,\ \ B_C + B_D \geq 6,\ \ B_D \geq 3,\ \ \ E_j, B_j \geq 0
\end{aligned}$$

2. Optimal Solutions (see `P3-21.xls`, Sheets (a) and (b))

- **Part (a): 44 agents total**: $(E_A, E_B, E_C, E_D) = (6, 7, 9, 9)$ and $(B_A, B_B, B_C, B_D) = (6, 1, 3, 3)$.
- **Part (b): 44 agents total**: $(E_A, E_B, E_C, E_D) = (12, 3, 9, 9)$ and $(B_A, B_B, B_C, B_D) = (4, 1, 3, 3)$, with minimum wage cost **\$880.00/hr** (or \$5,280.00/day). The total of 44 agents does *not* change.

Spreadsheet Row	**Eng A**	**Eng B**	**Eng C**	**Eng D**	**Bil A**	**Bil B**	**Bil C**	**Bil D**	**LHS (b)**	**Sign**	**RHS**
Part (a) Assigned	6.0	7.0	9.0	9.0	6.0	1.0	3.0	3.0	**44.0**	*(Min Agents)*	
Part (b) Assigned	12.0	3.0	9.0	9.0	4.0	1.0	3.0	3.0			
Part (b) Pay (\$/hr)	19	19	19	19	23	23	23	23	**\$880.00**	*(Min Pay/hr)*	
6am–9am / 9am–12n	1, 1	0, 1	0, 0	0, 0	1, 1	0, 1	0, 0	0, 0	16, 20	$\geq$	12, 20
12n–3pm / 3pm–6pm	0, 0	1, 0	1, 1	0, 1	0, 0	1, 0	1, 1	0, 1	16, 24	$\geq$	16, 24
6pm–9pm needs	0	0	0	1	0	0	0	1	12.0	$\geq$	12.0
English (5 periods)	1	1	1	1	0	0	0	0	12, 15, 12, 18, 9	$\geq$	6, 10, 8, 12, 6
Bilingual (5 periods)	0	0	0	0	1	1	1	1	4, 5, 4, 6, 3	$\geq$	3, 5, 4, 6, 3

Problem 3-29: Margaret Young's Family Farm

Margaret Young's family owns five parcels of farmland, broken into a southeast sector, a north sector, a northwest sector, a west sector, and a southwest sector. Young is involved primarily in growing wheat, alfalfa, and barley crops and is currently preparing her production plan for next year. The Pennsylvania Water Authority has just announced its yearly water allotment, with the Young farm receiving 7,400 acre-feet. Each parcel can only tolerate a certain amount of irrigation per growing season, as specified in the following table:

Parcel	Area (Acres)	Water Limit (Acre-Ft)
Southeast	2,000	3,200
North	2,300	3,400
Northwest	600	800
West	1,100	500
Southwest	500	600

Crop	Maximum Sales	Water/Acre
Wheat	110,000 bushels	1.6
Alfalfa	1,800 tons	2.9
Barley	2,200 tons	3.5

Each of Young's crops needs a minimum amount of water per acre, and there is a projected limit on sales of each crop, as shown in the second table above. Young's best estimate is that she can sell wheat at a net profit of \$2 per bushel, alfalfa at \$40 per ton, and barley at \$50 per ton. One acre of land yields an average of 1.5 tons of alfalfa and 2.2 tons of barley. The wheat yield is approximately 50 bushels per acre. Formulate and solve Young's production plan as an LP model.

Solution to Problem 3-29

1. Decision Variables and Parameter Conversions

Let X_{ij} = acres of crop $i \in \{1 \text{ (Wheat)}, 2 \text{ (Alfalfa)}, 3 \text{ (Barley)}\}$ planted on parcel $j \in \{1 \text{ (SE)}, 2 \text{ (N)}, 3 \text{ (NW)}, 4 \text{ (W)}, 5 \text{ (SW)}\}$. Converting yields and sales limits to a per-acre basis:

- **Wheat ($i = 1$):** Profit = \$2/bu × 50 bu/acre = \$100/acre; Max acreage = 110,000/50 = 2,200 acres
- **Alfalfa ($i = 2$):** Profit = \$40/ton × 1.5 tons/acre = \$60/acre; Max acreage = 1,800/1.5 = 1,200 acres
- **Barley ($i = 3$):** Profit = \$50/ton × 2.2 tons/acre = \$110/acre; Max acreage = 2,200/2.2 = 1,000 acres

2. Linear Programming Formulation

$$\text{Maximize Profit} = \$100 \sum_{j=1}^{5} X_{1j} + \$60 \sum_{j=1}^{5} X_{2j} + \$110 \sum_{j=1}^{5} X_{3j}$$

subject to:

$$\textbf{Parcel Acreage:} \quad \sum_{i=1}^{3} X_{i1} \le 2{,}000, \ \sum_{i=1}^{3} X_{i2} \le 2{,}300, \ \sum_{i=1}^{3} X_{i3} \le 600, \ \sum_{i=1}^{3} X_{i4} \le 1{,}100, \ \sum_{i=1}^{3} X_{i5} \le 500$$

$$\textbf{Crop Sales (acres):} \quad \sum_{j=1}^{5} X_{1j} \le 2{,}200 \text{ (Wheat)}, \ \sum_{j=1}^{5} X_{2j} \le 1{,}200 \text{ (Alfalfa)}, \ \sum_{j=1}^{5} X_{3j} \le 1{,}000 \text{ (Barley)}$$

$$\textbf{Parcel Water:} \quad 1.6X_{1j} + 2.9X_{2j} + 3.5X_{3j} \le W_j \quad \text{for } (W_1, \ldots, W_5) = (3{,}200,\ 3{,}400,\ 800,\ 500,\ 600)$$

$$\textbf{Total Water:} \quad \sum_{j=1}^{5} (1.6X_{1j} + 2.9X_{2j} + 3.5X_{3j}) \le 7{,}400 \text{ acre-feet}, \qquad X_{ij} \ge 0 \ \ \forall i, j$$

3. Optimal Solution (see `P3-29.xls`)

Multiple optimal solutions exist, all yielding a maximum profit of \$**337,862.07** by planting 2,200 acres of Wheat, 1,000 acres of Barley, and 131.03 acres of Alfalfa:

Crop	SE	North	NW	West	SW	Total Acres	Sales Limit
Wheat (\$100/acre)	1,762.50	437.50	0.00	0.00	0.00	2,200.00	≤ 2,200
Alfalfa (\$60/acre)	131.03	0.00	0.00	0.00	0.00	131.03	≤ 1,200
Barley (\$110/acre)	0.00	771.43	228.57	0.00	0.00	1,000.00	≤ 1,000
Acres Used / Limit	1,893.53/2,000	1,208.93/2,300	228.57/600	0/1,100	0/500	**3,331.03**	
Water Used / Limit	3,200/3,200	3,400/3,400	800/800	0/500	0/600	**7,400.00**	≤ 7,400

Problem 3-34: Battery Park Stable Horse Feed

Battery Park Stable feeds and houses the horses used to pull tourist-filled carriages through the streets of Charleston's historic waterfront area. The stable owner, a former racehorse trainer, recognizes the need to set a nutritional diet for the horses in his care. At the same time, he would like to keep the overall daily cost of feed to a minimum. The feed mixes available for the horses' diet are an oat product, a highly enriched grain, and a mineral product. Each of these mixes contains a certain amount of five ingredients needed daily to keep the average horse healthy. The following table shows these minimum requirements, units of each nutrient per pound of feed mix, and costs for the three mixes:

	Feed Mix			
Nutrient	**Oat (Units/lb)**	**Grain (Units/lb)**	**Mineral (Units/lb)**	**Min. Needed (Units/day)**
A	2.0	3.0	1.0	6
B	0.5	1.0	0.5	2
C	3.0	5.0	6.0	9
D	1.0	1.5	2.0	8
E	0.5	0.5	1.5	5
Cost/lb	\$0.09	\$0.14	\$0.17	

In addition, the stable owner is aware that an overfed horse is a sluggish worker. Consequently, he determines that 5 pounds of feed per day is the most that any horse needs to function properly. Formulate this problem and solve for the optimal daily mix of the three feeds.

Solution to Problem 3-34

1. Decision Variables

Let A = pounds of oat product, G = pounds of enriched grain, and M = pounds of mineral product fed per horse each day.

2. Linear Programming Formulation

$$\text{Minimize Daily Cost} = \$0.09A + \$0.14G + \$0.17M$$

subject to:

$2.0A + 3.0G + 1.0M \geq 6$ (Nutrient A), $0.5A + 1.0G + 0.5M \geq 2$ (Nutrient B)

$3.0A + 5.0G + 6.0M \geq 9$ (Nutrient C), $1.0A + 1.5G + 2.0M \geq 8$ (Nutrient D)

$0.5A + 0.5G + 1.5M \geq 5$ (Nutrient E), $A + G + M \leq 5$ (Max feed/day, lbs), $A, G, M \geq 0$

3. Optimal Solution (see `P3-34.xls`)

- **Oat product (A):** 1.333 lbs ($1\frac{1}{3}$ lbs), **Enriched grain (G):** 0.000 lbs, **Mineral product (M):** 3.333 lbs ($3\frac{1}{3}$ lbs)
- **Minimum Daily Feed Cost:** \$0.69 (\$0.6867) per horse

Spreadsheet Row	**Oat (A)**	**Grain (G)**	**Mineral (M)**	**LHS**	**Sign**	**RHS**
Number of pounds	1.333	0.000	3.333			
Cost (\$)	0.09	0.14	0.17	**\$0.6867**	*(Objective)*	
Nutrient A	2.0	3.0	1.0	6.000	$\geq$	6.0
Nutrient B	0.5	1.0	0.5	2.333	$\geq$	2.0
Nutrient C	3.0	5.0	6.0	24.000	$\geq$	9.0
Nutrient D	1.0	1.5	2.0	8.000	$\geq$	8.0
Nutrient E	0.5	0.5	1.5	5.667	$\geq$	5.0
Max feed needed	1.0	1.0	1.0	4.667	$\leq$	5.0

Problem 3-35: Campus Dietitian Meal Plan

Kathy Roniger, campus dietitian for a small Idaho college, is responsible for formulating a nutritious meal plan for students. For an evening meal, she feels that the following five meal-content requirements should be met: (1) between 900 and 1,500 calories; (2) at least 4 milligrams of iron; (3) no more than 50 grams of fat; (4) at least 26 grams of protein; and (5) no more than 50 grams of carbohydrates. On a particular day, Roniger's food stock includes seven items that can be prepared and served for supper to meet these requirements. The cost per pound for each food item and its contribution to each of the five nutritional requirements are given in the following table:

	Table of Food Values* and Costs					
Food Item	**Calories/lb**	**Iron (mg/lb)**	**Fat (gm/lb)**	**Protein (gm/lb)**	**Carbs (gm/lb)**	**Cost/lb**
Milk	295	0.2	16.0	16	22	\$0.60
Ground meat	1,216	0.2	96.0	81	0	\$2.35
Chicken	394	4.3	9.0	74	0	\$1.15
Fish	358	3.2	0.5	83	0	\$2.25
Beans	128	3.2	0.8	7	28	\$0.58
Spinach	118	14.1	1.4	14	19	\$1.17
Potatoes	279	2.2	0.5	8	63	\$0.33

*Source: C. F. Church and H. N. Church, *Bowes and Church's Food Values of Portions Commonly Used*, 12/e. Philadelphia: J.B. Lippincott, 1975.

(a) What combination and amounts of food items will provide the nutrition Roniger requires at the least total food cost? Formulate and solve this as an LP problem.

(b) Is this a well-balanced diet? Why or why not?

Solution to Problem 3-35

(a) Decision Variables and LP Formulation

Let $M, G, C, F, B, S, P \geq 0$ be the pounds of milk, ground meat, chicken, fish, beans, spinach, and potatoes included in the meal:

$$\text{Minimize Total Cost} = \$0.60M + \$2.35G + \$1.15C + \$2.25F + \$0.58B + \$1.17S + \$0.33P$$

subject to:

$$\begin{aligned}
900 \leq 295M + 1{,}216G + 394C + 358F + 128B + 118S + 279P \leq 1{,}500 \quad &\text{(Calories)} \\
0.2M + 0.2G + 4.3C + 3.2F + 3.2B + 14.1S + 2.2P \geq 4 \quad &\text{(Min iron, mg)} \\
16.0M + 96.0G + 9.0C + 0.5F + 0.8B + 1.4S + 0.5P \leq 50 \quad &\text{(Max fat, gm)} \\
16M + 81G + 74C + 83F + 7B + 14S + 8P \geq 26 \quad &\text{(Min protein, gm)} \\
22M + 28B + 19S + 63P \leq 50 \quad &\text{(Max carbs, gm)}
\end{aligned}$$

Optimal Solution (see `P3-35.xls`):

- **Ground meat (G):** 0.499 lbs, **Chicken (C):** 0.173 lbs, **Spinach (S):** 0.105 lbs, **Potatoes (P):** 0.762 lbs ($M = F = B = 0$); **Min Cost:** **\$1.75** (\$1.7460)

Row	**Milk**	**Ground**	**Chicken**	**Fish**	**Beans**	**Spinach**	**Potatoes**	**LHS**	**Sign**	**RHS**
Pounds	0.000	0.499	0.173	0.000	0.000	0.105	0.762			
Cost (\$)	0.60	2.35	1.15	2.25	0.58	1.17	0.33	**\$1.75**	*(Objective)*	
Min calories	295	1,216	394	358	128	118	279	900.00	$\geq$	900
Min iron	0.2	0.2	4.3	3.2	3.2	14.1	2.2	4.00	$\geq$	4
Min protein	16	81	74	83	7	14	8	60.78	$\geq$	26
Max calories	295	1,216	394	358	128	118	279	900.00	$\leq$	1,500
Max fat	16	96	9	0.5	0.8	1.4	0.5	50.00	$\leq$	50
Max carbs	22	0	0	0	28	19	63	50.00	$\leq$	50

(b) Evaluation of the Diet: The meal is fairly well-balanced (two meats, a green vegetable, and a potato), and the weight of each item is realistic.

Problem 3-38: Olive Oil Blending

A distributor imports olive oil from Spain and Italy in large casks. He then mixes these oils in different proportions to create three grades of olive oil that are sold domestically in the United States. The domestic grades include (a) Commercial, which must be no more than 35% Italian, (b) Virgin, which may be any mix of the two olive oils, and (c) Extra Virgin, which must be at least 55% Spanish. The cost to the distributor for Spanish olive oil is \$6.50 per gallon. Italian olive oil costs him \$5.75 per gallon. The weekly demand for the three types of olive oils is 700 gallons of Commercial, 2,200 gallons of Virgin, and 1,400 gallons of Extra Virgin.

How should he blend the two olive oils to meet his demand most economically?

Solution to Problem 3-38

1. Decision Variables

Let the six oil-to-blend quantities (in gallons) be:

- I_C, I_V, I_E = gallons of Italian olive oil blended into Commercial, Virgin, and Extra Virgin, respectively
- S_C, S_V, S_E = gallons of Spanish olive oil blended into Commercial, Virgin, and Extra Virgin, respectively

2. Linear Programming Formulation

$$\text{Minimize Total Cost} = \$5.75(I_C + I_V + I_E) + \$6.50(S_C + S_V + S_E)$$

subject to:

$$\begin{aligned}
I_C + S_C &\geq 700 && \text{(Commercial demand, in gallons)}\\
I_V + S_V &\geq 2{,}200 && \text{(Virgin demand, in gallons)}\\
I_E + S_E &\geq 1{,}400 && \text{(Extra Virgin demand, in gallons)}\\
I_C \leq 0.35(I_C + S_C) &\iff 0.65I_C - 0.35S_C \leq 0 && \text{(Max 35\% Italian in Commercial)}\\
S_E \geq 0.55(I_E + S_E) &\iff -0.55I_E + 0.45S_E \geq 0 && \text{(Min 55\% Spanish in Extra Virgin)}\\
I_C,\, I_V,\, I_E,\, S_C,\, S_V,\, S_E &\geq 0 && \text{(Non-negativity)}
\end{aligned}$$

3. Optimal Solution (see `P3-38.xls`)

- **Commercial (700 gal):** $I_C = 245$ gal Italian (35%), $S_C = 455$ gal Spanish (65%)
- **Virgin (2,200 gal):** $I_V = 2{,}200$ gal Italian (100%), $S_V = 0$ gal Spanish (0%)
- **Extra Virgin (1,400 gal):** $I_E = 630$ gal Italian (45%), $S_E = 770$ gal Spanish (55%)
- **Minimum Weekly Cost:** \$25,643.75

Row	Ital Comm	Ital Virg	Ital Extra	Span Comm	Span Virg	Span Extra	LHS	Sign	RHS
Gallons	245.0	2,200.0	630.0	455.0	0.0	770.0			
Cost (\$)	5.75	5.75	5.75	6.50	6.50	6.50	**\$25,643.75** *(Obj.)*		
Comm. demand	1	0	0	1	0	0	700.0	≥	700.0
Virgin demand	0	1	0	0	1	0	2,200.0	≥	2,200.0
Extra demand	0	0	1	0	0	1	1,400.0	≥	1,400.0
Min Span Extra	0	0	0	0	0	1	770.0	≥	770.0
Max Ital Comm	1	0	0	0	0	0	245.0	≤	245.0

Problem 3-42: South Central Utilities Nuclear Staffing

South Central Utilities has just announced the August 1 opening of its second nuclear generator at its Baton Rouge, Louisiana, nuclear power plant. Its personnel department has been directed to determine how many nuclear technicians need to be hired and trained over the remainder of the year. The plant currently employs 350 fully trained technicians and projects the following personnel needs:

Month	August	September	October	November	December
Personnel Hours Needed	40,000	45,000	35,000	50,000	45,000

By Louisiana law, a reactor employee can actually work no more than 130 hours per month. (Slightly over 1 hour per day is used for check-in and check-out, record keeping, and daily radiation health scans.) Policy at South Central Utilities also dictates that layoffs are not acceptable in those months when the nuclear plant is overstaffed. So, if more trained employees are available than are needed in any month, each worker is still fully paid, even though he or she is not required to work the 130 hours.

Training new employees is an important and costly procedure. It takes one month of one-on-one classroom instruction before a new technician is permitted to work alone in the reactor facility. Therefore, South Central must hire trainees one month before they are actually needed. Each trainee teams up with a skilled nuclear technician and requires 90 hours of that employee's time, meaning that 90 hours less of the technician's time are available that month for actual reactor work.

Personnel department records indicate a turnover rate of trained technicians at 2% per month. In other words, 2% of the skilled technicians at the start of any month resign by the end of that month. A trained technician earns a monthly salary of \$4,500, while trainees are paid \$2,000 during their one month of instruction. Formulate this staffing problem using LP and solve it by using Excel.

Solution to Problem 3-42

1. Decision Variables and Formulation

For month $i \in \{1\text{ (Aug)}, 2\text{ (Sept)}, 3\text{ (Oct)}, 4\text{ (Nov)}, 5\text{ (Dec)}\}$, let X_i = trained technicians available at the start of month i, and Y_i = trainees hired at the start of month i.

$$\text{Minimize Total Payroll} = \$4{,}500(X_1 + X_2 + X_3 + X_4 + X_5) + \$2{,}000(Y_1 + Y_2 + Y_3 + Y_4 + Y_5)$$

subject to:

$$130X_1 - 90Y_1 \geq 40{,}000 \text{ (Aug)}, \qquad 130X_2 - 90Y_2 \geq 45{,}000 \text{ (Sept)}, \qquad 130X_3 - 90Y_3 \geq 35{,}000 \text{ (Oct)}$$
$$130X_4 - 90Y_4 \geq 50{,}000 \text{ (Nov)}, \qquad 130X_5 - 90Y_5 \geq 45{,}000 \text{ (Dec)}, \qquad X_1 = 350 \text{ (Aug 1)}$$
$$-0.98X_i + X_{i+1} - Y_i = 0 \quad (i = 1, 2, 3, 4), \qquad X_i,\, Y_i \geq 0 \quad (i = 1, \ldots, 5)$$

2. Optimal Solution (see `P3-42.xls`)

- **Trained Technicians ($X_1, \ldots, X_5$):** Aug = 350.00, Sept = 346.15, Oct = 339.23, Nov = 384.62, Dec = 376.92
- **Trainees Hired ($Y_1, \ldots, Y_5$):** Aug = 3.15, Sept = 0.00, Oct = 52.17, Nov = 0.00, Dec = 0.00; **Payroll: \$8,196,800**

Row	T. Aug	T. Sep	T. Oct	T. Nov	T. Dec	Tr. Aug	Tr. Sep	Tr. Oct	Tr. Nov	Tr. Dec	LHS	Sign	RHS
People	350.00	346.15	339.23	384.62	376.92	3.15	0.00	52.17	0.00	0.00			
Salary	4,500	4,500	4,500	4,500	4,500	2,000	2,000	2,000	2,000	2,000	**\$8,196,800**	*(Obj.)*	
Aug need	130	0	0	0	0	−90	0	0	0	0	45,216.15	≥	40,000
Sept need	0	130	0	0	0	0	−90	0	0	0	45,000.00	≥	45,000
Oct need	0	0	130	0	0	0	0	−90	0	0	39,404.77	≥	35,000
Nov need	0	0	0	130	0	0	0	0	−90	0	50,000.00	≥	50,000
Dec need	0	0	0	0	130	0	0	0	0	−90	49,000.00	≥	45,000
Aug tech	1	0	0	0	0	0	0	0	0	0	350.00	=	350
Sept tech	−0.98	1	0	0	0	−1	0	0	0	0	0.00	=	0
Oct tech	0	−0.98	1	0	0	0	−1	0	0	0	0.00	=	0
Nov tech	0	0	−0.98	1	0	0	0	−1	0	0	0.00	=	0
Dec tech	0	0	0	−0.98	1	0	0	0	−1	0	0.00	=	0

Problem 3-45: Mt. Sinai Hospital Expansion

Mt. Sinai Hospital in New Orleans is a large, private, 600-bed facility complete with laboratories, operating rooms, and x-ray equipment. In seeking to increase revenues, Mt. Sinai's administration has decided to make a 90-bed addition on a portion of adjacent land currently used for staff parking. The administrators feel that the labs, operating rooms, and x-ray department are not being fully utilized at present and do not need to be expanded to handle additional patients. The addition of 90 beds, however, involves deciding how many beds should be allocated to the medical staff for medical patients and how many to the surgical staff for surgical patients.

The hospital's accounting and medical records departments have provided the following pertinent information. The average hospital stay for a medical patient is 8 days, and the average medical patient generates \$2,280 in revenues. The average surgical patient is in the hospital 5 days and receives a \$1,515 bill. The laboratory is capable of handling 15,000 tests per year more than it has been handling. The average medical patient requires 3.1 lab tests, and the average surgical patient takes 2.6 lab tests. Furthermore, the average medical patient uses one x-ray, whereas the average surgical patient requires two x-rays. If the hospital was expanded by 90 beds, the x-ray department could handle up to 7,000 x-rays without significant additional cost. Finally, the administration estimates that up to 2,800 additional operations could be performed in existing operating room facilities. Medical patients, of course, require no surgery, whereas each surgical patient generally has one surgery performed.

Formulate this problem so as to determine how many medical beds and how many surgical beds should be added to maximize revenues. Assume that the hospital is open 365 days a year. Then solve the problem.

Solution to Problem 3-45

1. Decision Variables and Formulation

Let M = additional medical patients admitted per year, and S = additional surgical patients admitted per year. With 90 new beds open 365 days/year, the addition provides $90 \times 365 = 32{,}850$ patient-days per year:

$$\text{Maximize Annual Revenue} = \$2{,}280M + \$1{,}515S$$

subject to:

$$8M + 5S \leq 32{,}850 \text{ (Annual bed-days)}, \quad 3.1M + 2.6S \leq 15{,}000 \text{ (Lab tests/year)}$$
$$1.0M + 2.0S \leq 7{,}000 \text{ (X-rays/year)}, \quad 1.0S \leq 2{,}800 \text{ (Surgeries/year)}, \quad M, S \geq 0$$

2. Optimal Solution and Bed Allocation (see `P3-45.xls`)

- **Medical (M):** 2,790.91 patients/yr ($\approx$ 2,791), **Surgical (S):** 2,104.55 patients/yr ($\approx$ 2,105)
- **Maximum Annual Revenue:** \$9,551,659.09 ($\approx$ \$9,551,659)
- **Bed Allocation:** Medical = $(2{,}790.91 \times 8)/365 = 61.17 \approx$ **61** medical beds (68%); Surgical = $(2{,}104.55 \times 5)/365 = 28.83 \approx$ **29** surgical beds (32%).

Spreadsheet Row	Medical (M)	Surgical (S)	LHS	Sign	RHS
Number of patients	2,790.91	2,104.55			
Revenue (\$)	2,280.00	1,515.00	**\$9,551,659.09**	*(Objective)*	
Patient-days	8.0	5.0	32,850.00	$\leq$	32,850.00
Lab tests	3.1	2.6	14,123.64	$\leq$	15,000.00
X-rays	1.0	2.0	7,000.00	$\leq$	7,000.00
Oper/Surgeries	0.0	1.0	2,104.55	$\leq$	2,800.00